The Structural Theory of Intelligence Under Bounded Context

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Abstract

A companion paper [McCarthy2026a] establishes that any system accumulating propositions under bounded active context C_n must maintain a directed graph with typed edges — graph structure is not a design choice but a necessary consequence of information preservation. This paper derives the complete structural theory of intelligence that follows from that foundation.

Starting from bounded C_n, a survival threshold U^d_{lethal}, and a world with positive entropy, we prove a connected chain of necessary consequences: knowledge and action are inseparable (K is constitutively indexed by the informed action space F, with an asymmetric retention criterion derived from survival pressure); the escalation architecture dominates top-down allocation (bottom-up failure signaling is $O(1)$ in steady state; precision allocation degrades as $O(|F|/C_n)$); uncertainty reduction follows evaluation-driven state revision (gradient descent on U with convergence guaranteed by a scalar Lyapunov argument); the encoding hierarchy follows from cost structure (proven actions promote to lower levels with rigor threshold $\theta_i = 1 - \varepsilon/C_i$); F and M are type-irreducible; and agency $\mathcal{A} > 0$ and learning efficiency $\eta_M > 0$ are the unique structural invariants of any persisting entity.

Intelligence is therefore the **driven capacity to improve**: $I(E) = \mathcal{A} \cdot \eta_M$ — not what has been learned, but how efficiently evidence converts into better (F, K). This is not stipulated but derived: $\mathcal{A}$ and η_M are the only properties that every persisting entity must have, and their product is the unique normalized combination of a probability and a conditional rate. Extensions to collective intelligence, sentience, and superintelligence are developed in a separate companion paper.

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1. Introduction

From graph necessity to the structural theory. A companion paper [McCarthy2026a] proves that any system accumulating propositions under bounded active context must maintain a directed graph with typed edges over its propositions. The proof is substrate-independent: it holds for biological neural networks, artificial systems, and institutional knowledge alike. The present paper takes that result as a foundation and asks: *given that graph structure is necessary, what is the complete set of structural consequences for intelligence?*

The answer is a connected chain. Starting from bounded context C_n, a survival

threshold U^d_{lethal}, and a world with positive entropy, we derive: the inseparability of knowledge and action (Theorem 1); the dominance of escalation over top-down allocation (Theorems 2, 2a); evaluation-driven state revision with guaranteed convergence (Theorem 3); the encoding hierarchy with rigor thresholds (Theorem 4); the type irreducibility of action and learning (Theorem 5); and the structural necessity of agency and learning efficiency (Theorem 6). Each result is a consequence of the same generative constraint. The individual results formalize known intuitions; the contribution is that they form a single deductive chain from one constraint.

The engineering origin. This framework emerged from a concrete engineering problem: generating repeatable, transposable tasks across a diverse codebase ecosystem spanning thousands of repositories. The formal theory describes what a self-optimizing agentic system converged to under real performance constraints. The biological and civilizational correspondences are predictions the framework makes — not the premises it rests on.

Provenance was the root constraint. An agentic system that synthesizes its behavior from a context window is inherently unpredictable. Transposability requires a quantifiable basis: evidence that an action has worked, attributed to a source, grounded in the conditions under which it succeeded. Without provenance — without knowing *why* something worked — you cannot measure confidence, and without a confidence measure you cannot optimize. The system converged on a knowledge graph separating *why you know something* from *how to apply it*: the graph structure whose necessity is formally established in [McCarthy2026a].

The engineering consequence was precisely measurable. Before separating K from F: $O(N \times M)$ retrieval cost with no sharing of common structure. After separation: $O(1)$ local graph traversal per entity, independent and unbounded. The $O(N \times M) \to O(1)$ transformation is the empirical measurement of K precipitating from F as the normalization of shared propositions (Corollary 2b). Adding agents introduced the escalation architecture: subproblems that previously required explicit orchestration became self-contained agents that escalated only on failure. Proven subnodes converged toward determinism and were scripted — promoted to a lower encoding level. The active inference load stayed bounded as the system's coverage of W expanded.

Biology as feasibility test. The test of whether this is a theory of software optimization or a theory of intelligence is whether the formal structure is consistent with biology — the only system where intelligence is unambiguously instantiated over evolutionary time. We claim that if the theory is correct, the biological architecture should be feasible under it: that genetic encoding of M rather than (K, F) is the predicted transmission mode (Section 11, Open Question 8); that the encoding hierarchy from reflex to active reasoning follows from the cost structure of Theorem 4; that cultural transmission emerges as a structural necessity when the heritable channel cannot carry individual (K, F) across generations. None of these were assumed. They follow from the same

optimization constraints that produced the software architecture.

Relationship to prior accounts. The framework defines a structural class: the set of all processes satisfying bounded active context, a survival threshold, and a growing reachable space. Biological evolution is one member; gradient descent in artificial systems is another; collective human knowledge transmission is another. The reason they all converge on graph-structured knowledge, escalation architectures, and provenance requirements is that these are necessary properties of any member of the class (Theorem 6 and [McCarthy2026a]).

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2. The Central Claims

We claim: **intelligence is the driven capacity to improve** — agency $\mathcal{A}$ exercised through a higher-order function space M that continuously refines both the knowledge graph K and the informed action space F. Formally: $I(E) = \mathcal{A} \cdot \eta_M$.

Three components are jointly necessary and none is sufficient alone:

- **Drive ($\mathcal{A}$):** the intrinsic orientation toward uncertainty reduction — the desire to know — without which M has no reason to engage the gradient
- **Mechanism (η_M):** the efficiency of M: how effectively it converts gradient signals into improved (F, K). Without $\eta_M > 0$, the entity cannot learn from evidence; it executes a fixed (F, K) no matter how motivated
- **Outcome:** the bounding of uncertainty through knowledge-grounded action — what $I(E)$ produces over time, not what it is. The size of K or F measures past intelligence; $I(E)$ measures the rate at which that past is being extended

$\mathcal{A} > 0$ and $\eta_M > 0$ are the only universal invariants of intelligence (Theorem 6). Every other aspect of E — F, K, substrate, sensing function — can differ arbitrarily. $\mathcal{A} = 0$ or $\eta_M = 0$ yields the automaton: executing what prior gradient descent encoded, but not itself descending.

The contribution. Prior theories identify individual aspects of intelligent systems: FEP derives uncertainty minimization from self-organization [Friston2010]; Hutter defines optimal behavior over computable environments [Hutter2005]; Chollet measures skill-acquisition efficiency [Chollet2019]; Simon establishes bounded rationality [Simon1955]; Dupoux, LeCun, and Malik [DupouxLeCun-Malik2026] propose a dual-system architecture motivated by cognitive science. Each takes its structural assumptions as given.

A companion paper [McCarthy2026a] identifies the first link in the chain: bounded active context C_n under information-preservation pressure necessarily produces graph-structured knowledge. This paper derives the remaining structural consequences as a connected chain from the same generative constraint — bounded C_n under survival pressure in a world with positive entropy: the

inseparability of knowledge and action (Theorem 1), escalation dominance over top-down allocation (Theorem 2a), evaluation-driven state revision (Theorem 3), the encoding hierarchy (Theorem 4), type irreducibility (Theorem 5), and the necessity of $\mathcal{A}$ and η_M (Theorem 6). The individual results formalize known intuitions; the contribution is that they are all consequences of the same constraint, derived rather than separately assumed.

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3. Formal Definitions

Definition 1 (World State Space). Let W be a measurable space of possible world states. At time t, the true state $w_t \in W$ is not directly observable by any entity.

Definition 2 (Knowledge Graph). A knowledge graph K is a directed graph where nodes are propositions about W and edges represent inferential or causal relationships between propositions. K is not a passive store; its retention criterion is established by Theorem 1.

Definition 2a (Bounded Knowledge Graph). A knowledge graph K has capacity K_{max}: the maximum number of propositions K can hold simultaneously, determined by the substrate's physical constraints. When $|K| < K_{max}$, K operates in the **growth regime**: propositions accumulate freely subject to the retention criterion of Theorem 1. When $|K| = K_{max}$, K operates in the **curation regime**: adding a new proposition requires removing an existing one. The two regimes have different retention criteria.

Definition 3 (Informed Action Space). Let $F \subseteq (K \times W_{obs} \to \Delta A)$ be a set of functions, where W_{obs} is an entity's observable projection of W and ΔA is a distribution over possible actions. Each element $f \in F$ has type $f : K \times W_{obs} \to \Delta A$. F is the complete set of informed actions available to an entity. An **informed action** $f \in F$ is an action grounded in K, validated against W, and attributed to its origins: one that knows why it works. An uninformed action (grounded in no K) is not an element of F. Knowledge with no action potential is not an element of F either; it is data. An informed action is specifically the coupling: knowledge that enables action, action that is justified by knowledge.

F carries graph structure. The elements of F are nodes; typed relations between them are edges. This structure is not a design choice — it is forced by the same constraints (retrieval $O(|F^{[f]}|)$, bounded C_n, deduplication) that force K to be a directed graph (Theorem 2b Part 3), and independently by the requirement for sub-linear retention evaluation with cascade (Theorem 2c). The proof is identical: any structure satisfying all constraints simultaneously is a directed graph. Edge types on F follow from the same evidence-grounded logic as edge types on K: hard dependencies between actions (feasibility constraints), probabilistic ordering evidence (soft gradient directions), and joint-execution

synergies (combined actions that outperform either alone).

F as a graph is discrete (nodes and typed edges); each individual $f \in F$ maps into the continuous probability simplex ΔA. Gradient descent in F is traversal on the discrete graph, directed by the continuous-valued $G(f', K)$ at each adjacent node. K and F are both directed graphs under bounded context and selection pressure, differing only in domain: K is what the entity knows; F is what it can do.

In general, K and F are distinct graphs with many-to-many coupling. In the degenerate case where every proposition is directly actionable, $F = K$. This is the limit of crystallization at the bottom of the encoding hierarchy: a reflex is simultaneously knowledge and action. $F = K$ holds locally in fully crystallized domains; $F \neq K$ persists globally because $\mathcal{T}_{reachable}$ grows without bound (Theorem 3, Part II).

Definition 3a (Action Feedback). For any $f \in F$ executed in world state w, the feedback $\phi(f, w) \in \Phi$ is the observable consequence of f in w: the signal W returns in response to the action. The updated knowledge after executing f and observing its feedback is:

$$K_f = K \cup \{p_f\}$$

where p_f is the proposition grounded by observing $\phi(f, w)$. The feedback $\phi(f, w)$ is the formal object underlying the gradient signal δ in Theorem 3: $\delta = \phi(f, w) - \mathbb{E}[\phi(f, \cdot) \mid K]$.

Definition 3b (Information Gain). The information gain of action f given current knowledge K is:

$$G(f, K) = \mathbb{E}_w[U(w, K) - U(w, K_f)] = H(W \mid K) - H(W \mid K,\, \phi(f, w)) = I(f\,;\, W \mid K)$$

$G(f, K) \geq 0$ by the data processing inequality [Cover2006]: conditioning on true information cannot increase conditional entropy. $G(f, K) = 0$ when f provides no new information about w given K. Corrupted feedback may increase U and falls outside this definition's scope.

Definition 4 (Entity). An entity $E = (K, F, \sigma, \mathcal{A})$ where $\sigma : W \to W_{obs}$ is a sensing function projecting the true world state into the entity's observable space. K and F evolve over time via M; their initial values (K_0, F_0) are inherited from L_0 encoding. The bootstrapping problem is resolved by L_0: the entity begins from the encoded product of gradients that preceded it. Throughout this paper K and F refer to current-time values; the subscript 0 denotes initial conditions.

Definition 5 (Uncertainty). The uncertainty of entity E about world state w is the surprisal of w given K [Shannon1948, Cover2006]:

$$U(w, K) = -\log P(w \mid K)$$

K is a directed graph of propositions interpreted as a Bayesian network [Pearl1988]. Each proposition $p \in K$ contributes a likelihood factor $\ell_p : W \to [0, 1]$. The graph's edge structure encodes conditional independence: propositions with no path between them are conditionally independent given shared ancestors. The posterior over world states given the full knowledge graph is:

$$P(w \mid K) \propto P_0(w) \cdot \prod_{p \in K} \ell_p(w)$$

This is the standard Bayes posterior obtained by conditioning P_0 on all propositions simultaneously. $P(w \mid K)$ is well-defined and normalizable for any proper prior P_0 and bounded likelihood functions ℓ_p. The theory is parametric in P_0 — it does not depend on a specific prior, only on its existence and well-formedness.

The expected uncertainty over all world states is the conditional entropy:

$$H(W \mid K) = \mathbb{E}_w[U(w, K)] = -\sum_w P(w \mid K) \log P(w \mid K)$$

The minimization imperative (Theorem 1) operates on this expected quantity; $U(w, K)$ is the pointwise uncertainty the entity faces when it encounters specific world state w.

Definition 5a (Survival Threshold). For entity E in domain $d \subseteq W$, there exists a threshold U^d_{lethal} such that:

$$U(w, K) > U^d_{lethal} \Rightarrow E \text{ does not survive in } d$$

Above this threshold, uncertainty is so high that the entity cannot act effectively enough to persist. U^d_{lethal} is not a choice or a preference; it is imposed by W.

Definition 6 (Agency). Agency $\mathcal{A} \in [0, 1]$ is the probability that entity E engages its learning mechanism M in response to a gradient signal from W: the response gate. It is not an external objective imposed on E, nor a property derived from F or K. It is constitutive: $\mathcal{A}$ is what makes F an active informed action space rather than a static structure. At $\mathcal{A} = 0$, F is a set of possible functions with no internal reason to engage W. At $\mathcal{A} = 1$, E engages on every gradient signal it encounters. Intermediate values are meaningful: the response probability can be higher or lower, shaped by architecture, encoding history, and the density of unresolved uncertainty in the entity's reachable W. The informal gloss — the desire to know [Schmidhuber2010], the intrinsic drive to reduce uncertainty — is a phenomenological description of this gate, not the definition.

$\mathcal{A}$ has an empirical operationalization:

$$\mathcal{A}(E) \propto \lim_{U^d_{lethal} \to \infty} \frac{dU}{dt}(E)$$

An entity with genuine $\mathcal{A} > 0$ continues engaging M when survival pressure is removed; an entity with $\mathcal{A} = 0$ stops. This isolates intrinsic engagement from survival-compelled updating and is measurable in any system where survival pressure can be varied independently of capacity.

$\mathcal{A}$ precipitates from M by the same normalization argument that precipitates K from F. An entity that embeds its directional drive in every application of M recomputes orientation at every step — the M-analog of the $O(n \cdot k)$ scaling failure. An entity that maintains $\mathcal{A}$ as the shared directional structure across all M-applications achieves $O(1)$ access to its drive. $\mathcal{A}$ is therefore the first factor to precipitate from M's cross product: the normalized orientation that every M-application shares, extracted for the same reason K is extracted from F.

Definition 7 (Intelligence). The intelligence of E is the *driven capacity to improve*: the exercise of $\mathcal{A}$ through M to refine F and K in response to gradient signals [Schmidhuber2010, Thrun1998]:

$$I(E) = \mathcal{A} \cdot \eta_M$$

where $\mathcal{A} \in [0, 1]$ is the intrinsic drive (Definition 6) and η_M is the efficiency of the higher-order function space M (Definition 11a): the rate at which M improves F's uncertainty-reduction capacity per unit of learning signal. This expression is stated here as a definition and established as necessary in Theorem 6: $\mathcal{A}$ and η_M are the unique invariants of persistence under bounded context and selection pressure, and their product is the minimal measure that captures both.

At $\mathcal{A} = 0$: $I(E) = 0$ — a capable but inert learning mechanism is not intelligence. At $\eta_M = 0$: $I(E) = 0$ — a motivated entity whose M cannot improve F from evidence is an automaton.

The expected uncertainty reduction along a trajectory is a **performance measure** of current F and K — the consequence of past M activity, not the definition of intelligence. Two entities starting from identical K and F but different $I(E)$ will diverge: the entity with higher η_M builds better F faster, and eventually dominates in U-reduction even if initially equal in performance.

Intelligence is not the size of K or F. K and F are what intelligence has built. Intelligence is the efficiency with which M continues building.

Note on named components. The components K, F, M, and $\mathcal{A}$ named here are dimensions that have precipitated from the entity's full cross-product state — call it K^n — at the individual learning level. The theory's general result is that dimensional precipitation must occur under the joint constraints: as K^n grows,

shared structure must be extracted and named or retrieval cost grows without bound. What dimensions precipitate is a function of the entity's W and its history; the theory constrains the process, not the outcome. K, F, M, and $\mathcal{A}$ are the dimensions that fall out in entities with active individual learning mechanisms — the class this paper studies. Other projections precipitate differently: biological L_0 encoding (DNA) does not separate into K, F, M, $\mathcal{A}$; it is a dense cross-product that carries all survival-relevant structure without naming any of it as a distinct component. The named components are an epistemological convenience grounded in the studied entity class, not a universal constraint of the theory.

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4. The Inseparability of Knowledge and Action

Lemma 1 (Proposition Absence Lethality). For any entity E and proposition $p \in K$ that grounds some $f \in F$ in domain $d \subseteq W$: removing p from K weakly increases uncertainty in d, and in the limiting case where p is the sole grounding for the critical action in d, the increase may cause $U(w, K \setminus \{p\}) > U^d_{lethal}$, at which point E does not survive in d (Definition 5a).

Proof. Conditional entropy is monotonically non-increasing in the conditioning set [Cover2006]: $H(W \mid K) \leq H(W \mid K \setminus \{p\})$ for any $p \in K$. Removing p weakly increases expected uncertainty. If p grounds the only $f \in F$ applicable in domain d, then $K \setminus \{p\}$ contains no action grounded in d: F has no mechanism to reduce $U(w, K \setminus \{p\})$ below its current level in that domain when $w \in d$ is encountered. By Definition 5a, if $U(w, K \setminus \{p\}) > U^d_{lethal}$ at the encountered w, E does not survive. Therefore the cost of absent p when needed, C_{absent}, is bounded below by entity non-survival in critical domains — which dominates all finite costs. □

Theorem 1 (K/F Inseparability). For any entity $E = (K, F, \sigma)$, the retention criterion for any proposition p is asymmetric:

- p is **retained** in K when $\exists\, f \in F$ applicable to p, or when the survival benefit of p is uncertain
- p is **pruned** from K only when it is certain that p contributes zero to $P(E, \tau)$ for any τ, i.e., when it is known that no $f \in F$ can be grounded in p, now or in any reachable region of W
- p is **displaced** in the curation regime ($|K| = K_{max}$) when a candidate proposition q satisfies $\mathbb{E}[G(f_q, K)] > \mathbb{E}[G(f_p, K)]$, where p is the proposition with minimum marginal value in K. Displacement requires only that q contributes more at the margin than p — not that p is useless. The asymmetric retention criterion governs which propositions enter; the comparative criterion governs which are displaced when the graph is full. Information gain G is submodular — the marginal value of p depends on what else is in K, so the greedy displacement criterion is evaluated at the margin given the current full K, not in isolation. This is the correct

procedure: the entity estimates value relative to what it already holds, not relative to an empty graph [Nemhauser1978].

The criterion is not symmetric. Uncertainty of benefit is not the same as certainty of no benefit. An entity with $\mathcal{A}$, whose $\mathcal{T}_{reachable}$ grows with every action taken, retains p whose value it cannot yet see, because the gradient may reach that territory. Pruning requires the same standard as certainty (Definition 10): the benefit must be known to be zero across sufficient variation in W.

Proof. Let $B(p, \tau)$ denote the contribution of p to $P(E, \tau)$ along trajectory τ, and let $\mathcal{T}_{reachable}$ be the set of trajectories reachable from the entity's current state. Define the option value of retaining p:

$$OV(p) = P(\exists\, \tau^* \in \mathcal{T}_{reachable} : B(p, \tau^*) > 0) \cdot \mathbb{E}[B(p, \tau^*) \mid B(p, \tau^*) > 0]$$

Let $c_{retain} > 0$ denote the finite maintenance cost of holding p in K (memory, retrieval overhead), and $c_{prune} \geq 0$ the cost of pruning p (processing, reorganization). Both are finite by assumption: any physical knowledge store incurs bounded costs per proposition. The expected net value of retention is $NV_{retain}(p) = OV(p) - c_{retain}$.

The expected net value of pruning is:

$$NV_{prune}(p) = -c_{prune} - P(\exists\, \tau^* : B(p, \tau^*) > 0) \cdot C_{absent}$$

where C_{absent} is the cost incurred when p is needed but absent. By Lemma 1, C_{absent} is bounded below by entity non-survival in critical domains, which dominates all finite costs: $C_{absent} \gg c_{retain}$.

Retention is selected for when $NV_{retain}(p) > NV_{prune}(p)$. Since $C_{absent} \gg c_{retain}$, this reduces to: retention holds whenever

$$P(\exists\, \tau^* \in \mathcal{T}_{reachable} : B(p, \tau^*) > 0) > 0$$

i.e., whenever survival benefit is uncertain. Even a small probability of needing p, multiplied by the cost of its absence, outweighs finite maintenance.

Pruning holds only when $P(\exists\, \tau^* : B(p, \tau^*) > 0) = 0$ — i.e., when it is certain that p contributes zero benefit across all reachable trajectories. Any positive probability keeps the retention condition active. The asymmetry is strict: uncertain benefit is sufficient for retention; certain zero benefit is necessary for pruning.

In the curation regime ($|K| = K_{max}$), adding q requires displacing $p^* = \arg\min_{p \in K} \mathbb{E}[G(f_p, K)]$. Displacement occurs when $\mathbb{E}[G(f_q, K)] > \mathbb{E}[G(f_{p^*}, K)]$: the asymmetric criterion applied to a choice between two propositions rather than

between one and nothing. By Theorem 3 Part II, $\mathcal{T}_{reachable}$ grows monotonically, keeping future needs uncertain unless benefit is certainly zero — preserving the asymmetry across the entity's lifetime. □

Correspondence. This formalizes Gibson's affordances [Gibson1979] and extends the information bottleneck [Tishby2011], which treats the retention threshold as a design parameter; here it is derived from survival pressure. The asymmetric criterion resolves an apparent asymmetry: an entity retains p beyond the certainty horizon because the benefit is uncertain, not known to be zero.

Corollary 1 (Indexed Knowledge). K is constitutively indexed by F. Two entities with identical σ but different F maintain different K even when perceiving identical world states.

Corollary 2 (Compactness). K is bounded in size by F: only propositions applicable to available informed actions are retained.

Corollary 2a (Curation and Forgetting). Under bounded K, forgetting is correct behavior: the curation regime displaces lower-value propositions as $\mathcal{T}_{reachable}$ evolves, ensuring K tracks the current frontier.

Corollary 2b (K as the Normalization of F — the Scaling Argument). An entity can maintain informed actions without a separately structured K, embedding all propositional knowledge implicitly in the parameters of each $f \in F$. This is a valid but scaling-limited architecture. Let $n = |F|$ and $k =$ average number of shared propositions per function. Without K: context per action is $O(k)$ (each f carries its full background); update cost when one proposition changes is $O(n)$ (every f encoding it must be individually revised); total cost scales as $O(n \cdot k)$. With K normalized out: context per action is $O(|K^{[f]}|)$ — the relevant subgraph retrieved by traversal (Theorem 2b); update cost per proposition is $O(1)$ — one update to K propagates to all f that reference it; total cost scales as $O(n + k)$.

The difference $O(n \cdot k)$ versus $O(n + k)$ is not a constant factor — it is the difference between a system whose context cost grows multiplicatively with shared knowledge complexity and one whose context cost stays bounded under growth of F. Under bounded C_n, the implicit-K architecture does not scale: every increase in $|F|$ increases required context per action proportionally to the shared background knowledge it carries. K precipitates from F as the normalization of shared propositions — the factoring-out of common structure across functions — because this is the only architecture that keeps C_n bounded as F grows. An entity without K is not wrong; it is limited to the regime where $n \cdot k$ fits in C_n. Selection pressure at scale removes such entities in favor of those whose K has crystallized — literally, the precipitation of shared knowledge out of the action space into its own normalized graph. This is the formal explanation of the $O(N \times M) \to O(1)$ empirical measurement described in Section 1: each entity's local graph traversal is $O(1)$ precisely because shared propositions have been factored out of F into K.

5. The Escalation Principle

Definition 8 (Encoding Hierarchy). The encoding hierarchy is a continuous two-dimensional space $\mathcal{E}(c, r)$ parameterized by runtime cost $c \geq 0$ and reversibility $r \in [0, 1]$, where increasing c corresponds to more expensive engagement and increasing r to more easily revised encodings. Consistent with the continuity principle: a continuous being moves through this space continuously; there is no categorical jump between encoding modes, only a gradient of cost and permanence [Baars1988, Dehaene2011, Kahneman2011].

Boundaries exist where the qualitative character of encoding changes, requiring information to be translated as it crosses. These boundaries are **interfaces**: formal structures through which signals pass between qualitatively distinct encoding regimes. The named levels $L_0, L_1, \ldots, L_n$ are the interfaces at the significant boundaries of $\mathcal{E}(c, r)$:

Interface	Boundary crossed	Runtime Cost	Reversibility
L_n	Active reasoning / novel uncertainty	O(compute)	Fully reversible
$\vdots$	Learned patterns, habits	Decreasing	Decreasing
L_1	Reflex / autonomic	≈ 0	Semi-permanent
L_0	Heritable encoding	0	Permanent within lineage

Escalation (Theorem 2) is the **upward interface protocol**: the specification of what form a failure signal takes when crossing a boundary from lower c to higher c. Promotion (Theorem 4) is the **downward interface protocol**: the condition under which an action may cross a boundary from higher c to lower c. The cost C_i in Theorem 4 is a property of the interface, not of the continuous space between interfaces: it is the cost of failure when an encoding error has crossed the boundary and propagates at the lower level's permanence.

The number of interfaces is substrate-determined — how many qualitative boundaries exist in the entity's physical $\mathcal{E}(c, r)$. The theory specifies the structural properties every interface must have (monotone cost, monotone rigor threshold, promotion criterion); it does not prescribe how many interfaces exist or what they are called. The named levels in the table above are empirical illustrations, not definitional. The continuous space between interfaces admits arbitrary encodings; the interfaces themselves are discrete because a boundary is a specific location, not a region.

Theorem 2 (Escalation). An informed action $f \in F$ is handled at the lowest encoding level L_i such that $U(w, K^{[f]}) \approx 0$ for the domain of w relevant to f, where $K^{[f]} \subseteq K$ is the knowledge subset applicable to f. Escalation to L_{i+1} occurs if and only if L_i fails to bound uncertainty.

Proof. By Definition 8, the cost of engaging level L_i exceeds the cost of engaging L_{i-1}: the encoding hierarchy is parameterized by increasing runtime cost c, so $C_{i+1} > C_i$ for all i. Selection pressure therefore favors handling f at the lowest level sufficient to bound uncertainty: engaging L_{i+1} when L_i suffices incurs unnecessary cost without reducing U further. An entity that routinely over-escalates incurs higher encoding costs without survival benefit and is selected against. Conversely, if L_i fails to bound $U(w, K^{[f]})$ — that is, $U(w, K^{[f]}) > 0$ beyond acceptable residual — then f is not handled at L_i: the failure propagates upward until some L_j $(j > i)$ contains the uncertainty or the entity is harmed (Definition 5a). The active context window L_n is the level of last resort by construction: it is the highest-cost, fully reversible level. □

Corollary 3 (Context Window as Frontier). L_n is the frontier of unbounded uncertainty, not the seat of intelligence. Its engagement signals failure in lower-level encodings.

Corollary 4 (Simultaneous Multi-Level Activity). All levels operate concurrently: L_n handles novel uncertainty while lower levels handle their domains independently.

Why the Graph: Progressive Disclosure and the Bounded Context Theorem

Definition 8a (Active Context Capacity and Inference Quality). Let C_n denote the capacity of L_n: the maximum number of propositions simultaneously active in context. C_n is bounded above by the substrate's physical constraints — but the effective capacity available for any given problem is also shaped by the structure of K. Graph-structured knowledge enables chunking: a single graph node can represent a cluster of propositions that co-activate as a unit, making multiple propositions available at the cost of one context slot. Expertise increases effective C_n within a domain by compressing frequently co-occurring propositions into retrievable chunks — not by expanding the substrate bound, but by reducing the number of slots required per unit of relevant knowledge. The substrate sets the ceiling; the structure of K determines how much of W that ceiling covers. Let $\mathrm{ctx}(L_n, t)$ be the set of propositions loaded into L_n at time t, with $|\mathrm{ctx}(L_n, t)| \leq C_n$.

The **inference quality** of L_n for problem f is the relevance density of its content:

$$\rho(L_n, f) = \frac{|K^{[f]}|}{|\mathrm{ctx}(L_n, t)|}$$

$\rho = 1$: L_n holds exactly what f requires — maximum signal, zero noise. $\rho \to 0$: L_n is dominated by content irrelevant to f — inference degrades. Every proposition in $\text{ctx}(L_n)$ that is not in $K^{[f]}$ is noise: it contributes variance to L_n's inference without contributing signal. Minimizing that noise is equivalent to minimizing $|\text{ctx}(L_n)| - |K^{[f]}|$.

Definition 8b (Delegation Overhead). Under a top-down allocation architecture, L_n directs which subproblems are handled at lower levels. For each active delegation, L_n must hold: the subproblem specification, the expected return signal, and sufficient context to integrate the result when it arrives. Let $\alpha > 0$ denote the minimum context cost per active delegation — the overhead top-down allocation imposes on L_n per managed subproblem, regardless of whether that subproblem is routine or novel.

Theorem 2a (Top-Down Allocation Degrades Inference Quality). Under a top-down allocation architecture with $k(t)$ concurrently delegated subproblems, the inference quality available to L_n for genuinely novel problems satisfies:

$$\rho(L_n, f_{\text{novel}}) = \frac{|K^{[f_{\text{novel}}]}|}{|K^{[f_{\text{novel}}]}| + k(t) \cdot \alpha}$$

As the entity's informed action space F grows (more capability, more concurrent processes), $k(t)$ grows proportionally and $\rho \to 0$. The only recovery is to grow C_n: the context window must scale linearly with capability to maintain fixed ρ.

Proof. L_n holds $K^{[f_{\text{novel}}]}$ for the current problem plus $k(t) \cdot \alpha$ for delegation overhead. Since $\alpha > 0$ and $k(t)$ grows with $|F|$, the denominator is unbounded while $|K^{[f_{\text{novel}}]}|$ remains bounded (any specific problem requires only its relevant subgraph). Therefore $\rho \to 0$ as $|F| \to \infty$. When $k(t) \cdot \alpha \geq C_n - |K^{[f_{\text{novel}}]}|$, L_n lacks capacity for genuine inference entirely. To maintain ρ at a fixed level, C_n must grow as $k(t)$ grows. □

Note on attention mechanisms. The $O(k)$ delegation overhead applies to any mechanism that evaluates all n stored propositions to determine relevance, including full self-attention ($O(n^2)$ per query). Sparse and learned attention mechanisms reduce this cost by learning fixed relevance patterns — which is learned graph construction. See Theorem 2c Part 6 for the full argument that continuous-weight retrieval mechanisms are implementations of graph structure, not alternatives to it.

Note. This is self-undermining on the Free Energy Principle's own terms [Friston2010, Friston2017]. FEP holds that intelligent systems minimize free energy — equivalently, minimize the variance they must manage. But growing C_n to accommodate delegation overhead adds variance: more propositions in L_n means more uncertain relevance to any specific problem, not less. The orchestrator architecture prescribed by FEP's precision allocation is inconsistent with FEP's own minimization imperative: allocating attention from L_n increases the variance that L_n must hold in order to allocate.

Biological note. We do not claim that biological precision allocation operates like a token context window, nor that the cortical hierarchy is unidirectional. Top-down signals exist at every level: higher levels necessarily convey signals to lower levels, and the bidirectionality of the cortical hierarchy is well established. The architectural claim is narrower: it concerns *engagement initiation*, not signal direction. In an escalation architecture, what triggers higher-level involvement is bottom-up failure — a lower level's inability to bound uncertainty — not a top-down assignment of work. Top-down signals flow as a consequence of engagement; they do not determine when engagement occurs. The cost of *top-down assignment as the primary control architecture* is $O(k)$ in high-level capacity, as the attentional and dual-task literature demonstrates — each concurrently supervised subprocess consumes a share of the supervisory resource. Escalation avoids this cost by making higher-level engagement demand-driven rather than supply-driven. The transition from active to automatic processing across skill acquisition is the biological measurement of this convergence: as f becomes automatic (assigned to L_0), it stops consuming high-level capacity, freeing L_n for novel problems. The formal argument is substrate-neutral; biology is the validation test.

Theorem 2b (Graph-Structured Knowledge Bounds Active Context). Under the escalation architecture (Theorem 2) with K structured as a directed graph (Definition 2), as K grows: (1) no delegation overhead at L_n; (2) inference quality $\rho = 1$ (conditional on edge fidelity); (3) C_n need not grow as $|F|$ or $|K|$ grows.

Proof summary. Parts 1–2 follow from escalation: L_n holds only the escalated problem's relevant subgraph $K^{[f]}$, not delegation context. Part 3 is graph necessity: the escalation architecture and bounded context jointly impose retrieval ($O(|K^{[f]}|)$), context ($\rho = 1$), and deduplication constraints on K, and graph structure is the only representation satisfying all three without information loss. The complete proof — including the information-preservation argument (factoring vs. discarding vs. lossy compression), the retention-dynamics argument (sub-linear dependency queries require stored adjacency), and the demonstration that attention mechanisms implement graph structure rather than constituting alternatives to it — is given in [McCarthy2026a]. Part 4: by Corollary 6, proven informed actions promote to lower encoding levels, keeping L_n occupied only by genuinely novel uncertainty. □

Corollary 4a (Selection Pressure Produces Graph-Structured Knowledge). Graph structure is not merely more efficient than alternatives — it is the necessary consequence of satisfying the retrieval, context, and memory constraints jointly [McCarthy2026a]. Entities that grow C_n to manage delegation overhead are selected against because they violate the context constraint as $|K|$ grows; entities using approximate retrieval are selected against because they violate $\rho = 1$; entities duplicating shared propositions are selected against because maintenance cost grows with $|F|$.

Corollary 4b (The Scaling Argument Against Context Window

Growth). Growing C_n directly is self-defeating [Liu2024]. By Theorem 2a, more context means more noise: ρ decreases as C_n fills with content not relevant to the current problem. By Definition 7, intelligence is the rate at which M improves (F, K), not the capacity of L_n. The efficient path is not to grow C_n but to grow the encoded knowledge at lower levels — filling the graph, not the context window.

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6. Gradient Descent on Uncertainty

Theorem 3 (Agency Necessarily Produces Evaluation-Driven State Revision).

On the space of descent and dimensionality. The gradient descent in this theorem operates in F — the space of informed actions — not in W. W may be arbitrarily rough, discontinuous, or of unknown dimensionality; no assumption on W's structure is required. The entity does not descend over W; it updates $f \in F$ in response to signals received from W.

The dimensionality of W or of M_{res} is not assumed. We do not know whether the latent space is finite or infinite-dimensional. The convergence proof does not require this knowledge. It operates on the scalar $U(w, K) = H(w \mid K)$, which is bounded below by zero regardless of the dimensionality of W or F. The proof shows $\mathbb{E}[U]$ decreases monotonically under M-updates and converges by the monotone convergence theorem, without decomposing U into independent dimensions.

Part I: Forced Signal, Response Gate. Every executed action f_i in world state w generates a feedback signal $\delta = \phi(f_i, w) - \mathbb{E}[\phi(f_i, \cdot) \mid K]$ (Definition 3a): the signed deviation of observed outcome from expectation. This signal arrives from W but is applied in F: it is a subgradient of U as a functional on F at the current f_i, evaluated at the encountered w. The signal is generated by execution, not by $\mathcal{A}$. It arrives whether or not the entity has $\mathcal{A}$. The entity does not choose whether to receive this signal. W imposes it.

$\mathcal{A}$ is the response gate: the probability that M engages on δ when δ arrives. An entity with $\mathcal{A} = 0$ receives every signal but M does not engage; (F, K) remains unchanged. An entity with $\mathcal{A} > 0$ engages M on the signal, updating (F, K) in response. The encountered world states are on-policy — determined by the entity's trajectory, not sampled i.i.d. from $P(w \mid K)$ — but this does not impede convergence. The subgradient δ is valid at every encountered w regardless of how w was reached; the entity is updating a function in F, and each δ is a legitimate descent direction in that space. The burn tells you that this f_i was insufficient at w.

An entity with $\mathcal{A} = 0$ does not update in response to δ. U does not decrease. It eventually exceeds U^d_{lethal} in some critical domain. Selection removes it. Therefore any entity that persists must have $\mathcal{A} > 0$: not because $\mathcal{A}$ generates

the signal, but because $\mathcal{A}$ is what enables the entity to act on signals that W imposes on every execution regardless. When M engages:

$$\Delta f_i = -\eta \cdot \delta$$

Convergence of U. The convergence argument does not decompose by dimension. $U(w, K) = H(w \mid K) \geq 0$ (Definition 5) is a scalar quantity bounded below by zero. We show it decreases in expectation under M-updates and use this to establish convergence directly.

Step 1 (Each update reduces U in expectation). By Definition 3b, every informed action $f \in F$ applied at world state w yields information gain $G(f, K) = I(f\,; w \mid K) \geq 0$, with equality only when f provides no new information about w given K. When M engages on the feedback signal δ and updates $(F, K) \to (F', K')$, the updated knowledge K' satisfies $H(w \mid K') \leq H(w \mid K)$ — conditioning on additional evidence cannot increase entropy (data processing inequality, [Cover2006]). Therefore:

$$\mathbb{E}[U(w, K') \mid K] \leq U(w, K) - G(f, K)$$

Each M-update that engages on a signal with $G > 0$ strictly reduces expected uncertainty. Updates with $G = 0$ leave U unchanged.

Step 2 (Convergence). The sequence $\{\mathbb{E}[U(w, K(t))]\}_{t \geq 0}$ is non-increasing (by Step 1) and bounded below by zero. By the monotone convergence theorem, $\mathbb{E}[U]$ converges. Since $\sum_t G(f_t, K(t)) \leq U(w, K(0)) < \infty$, the information gains $G(f_t, K(t)) \to 0$: the marginal improvement per update vanishes asymptotically. This is convergence to a local minimum of U given the entity's current F and σ.

Step 3 (Dimensionality is irrelevant to convergence). The argument above operates on the scalar U, not on individual dimensions of W or components of F. The entity's parameterization of F may grow as $\mathcal{T}_{reachable}$ expands and new propositions are grounded (Part II). This growth does not invalidate convergence: each new proposition p_{new} added to K can only reduce U (by the data processing inequality), and the cross-grounding of p_{new} with existing propositions can reduce U further by improving the resolution of knowledge already in K. Adding structure to K is analogous to adding a column to a matrix and re-normalizing: the existing entries can become better resolved, not worse. The coupling between new and existing knowledge *accelerates* convergence; it cannot prevent it, because the Lyapunov bound $U \geq 0$ holds regardless of dimensionality.

Part II: Non-termination. Let $\mathcal{T}_{reachable}(K)$ be the set of trajectories reachable from the entity's current state given knowledge K (as defined in Theorem 1). We claim that for any finite $K(t)$, the frontier of $\mathcal{T}_{reachable}$ contains world states w where $H(w \mid K) > 0$, provided W is sufficiently rich that each executed action can ground a genuinely new proposition. In a finite W where all propositions

are exhaustible, the frontier could in principle be covered; the claim applies in any W where the entity's epistemic horizon has not been closed — which is the intended domain.

By Definition 3a, executing any $f \in F$ produces feedback $\phi(f, w)$ that grounds a new proposition p_f not previously in K, yielding $K_f = K \cup \{p_f\}$. This richer K_f grounds new informed actions not available from K alone (Definitions 3 and 3b): actions that require p_f to select or apply, reaching world states not reachable without p_f. Therefore $\mathcal{T}_{reachable}(K_f) \supseteq \mathcal{T}_{reachable}(K)$, strictly whenever p_f grounds at least one new $f' \in F$. Since Part I establishes that every executed action generates a feedback signal, $\mathcal{T}_{reachable}$ grows monotonically with K. At any finite time t, $K(t)$ has not covered $\mathcal{T}_{reachable}(K(t))$: the frontier always contains states w where $H(w \mid K) > 0$. Therefore $\nabla_F U$ is never globally zero over $\mathcal{T}_{reachable}$, and descent never terminates within the entity's reachable epistemic domain.

Proof. Part I: $U \geq 0$ is a Lyapunov function; each M-update with $G > 0$ strictly reduces $\mathbb{E}[U]$ (data processing inequality); $\mathbb{E}[U]$ converges by monotone convergence. Entities with $\mathcal{A} = 0$ do not engage M and are removed by selection. Part II: follows from the monotonic growth of $\mathcal{T}_{reachable}$ with K — each executed action extends the reachable frontier, ensuring new unexplored states are always accessible within the domain of the entity's possible experience. □

Corollary 5 (Evaluation-Driven State Revision is Survival, Not Metaphor). For individual learning, gradient descent on U is not a description of how learning happens to work. It is what survival-driven updating *is*: the entity that burns and does not become more careful does not survive. The entity that survives is, by definition, the one whose M engaged on the signal. At the evolutionary timescale, the mechanism differs in kind: random mutation perturbs the L_0 encoding, generating new $(F, K, M, \mathcal{A})$ configurations; W evaluates each through the survival criterion; configurations that persist reproduce. No individual organism follows a gradient — the gradient is revealed by differential survival of randomly generated candidates. This is structurally the same as a locally irrational action (temporarily negative η_M) at the individual level: a perturbation that generates a new starting point for the $M \times \mathcal{A} \times K$ cross-product to be evaluated by W. The two mechanisms occupy a spectrum: at one end, random mutation with binary survival feedback (evolution); at the other, directed update via continuous δ carrying full gradient information (individual learning). Gradient descent is the efficient limit of this spectrum under two conditions: (i) δ carries directional information in F — not merely pass/fail, but a signed deviation identifying which aspect of f_i was insufficient at w; and (ii) M can use that directional information to bias the update toward lower U rather than generating a random candidate. When both conditions hold, each evaluation step directly reduces U. When neither holds, the mechanism is random search with W-evaluation. Real learning systems occupy positions between these limits; signal informativeness determines how closely the process approximates directed gradient descent.

Remark (Multi-timescale Structure of Evaluation-Driven Revision). Individual learning and evolutionary selection share a common structure: execute, receive signal, retain or revise. The mechanisms and mathematical objects differ in kind.

For individual learning, the mechanism is directed: the gradient signal δ arrives from a specific failure, M updates (F, K) in the direction that reduces U:

$$\Delta F_{\text{individual}} = -\eta_{\text{ind}} \nabla_F U(w, K)$$

For evolutionary selection, the mechanism is undirected: random mutation perturbs the L_0 encoding of $(F, K, M, \mathcal{A})$, generating candidate configurations. W evaluates each through survival. The aggregate population-level distribution shifts toward configurations with lower $\mathbb{E}[U]$, but no individual organism descends a gradient — the gradient is revealed by differential survival of randomly generated candidates. The notation $\Delta F_{\text{species}}$ denotes the shift in the population distribution over F-variants, not a gradient step by any individual:

$$\Delta F_{\text{species}} = \text{select}(\text{mutate}(F_{\text{population}}))$$

Both processes are instances of the same structure: generate a candidate state, submit to W's evaluation, retain if valid. Gradient descent ($\Delta F_{\text{individual}}$) is the efficient limit when signals are informative enough to direct updates. Mutation followed by selection ($\Delta F_{\text{species}}$) is the same mechanism without the directed signal. This structural equivalence is not an identity — mechanisms, timescales, and mathematical objects are all distinct — but it grounds the common observation that both processes move toward lower U at their respective timescales, and grounds a provable consequence in Part II.

Derivation.

Part I (Common structure, different mechanisms). Individual learning updates F for a single entity based on $U(w, K)$ at specific world states w encountered by that entity. Evolutionary change operates on a different object: it updates the population-level distribution over F-variants through differential reproduction of randomly generated candidates. These are not the same mathematical operation — individual learning changes a function within one agent via directed update; evolutionary change changes a probability distribution over many agents via undirected mutation and selection. The common structure is: generate a new internal state, submit it to W's evaluation, retain if valid. Gradient descent is the efficient limit of this structure; random mutation followed by selection is the general case.

Part II ($\eta_{\text{evo}} \ll \eta_{\text{ind}}$ follows from Theorem 4). Although the mechanisms differ, both rates can be characterized by the rigor threshold of their respective encoding levels. The learning rate η_i at encoding level L_i is inversely proportional

to the rigor threshold θ_i: a higher threshold requires more evidence per unit update, which is the definition of a lower update rate. Formally:

$$\eta_i \propto \frac{1}{\theta_i} = \frac{1}{1 - \varepsilon/C_i} = \frac{C_i}{C_i - \varepsilon}$$

For individual learning at L_n: C_n is the cost of a single wrong action, relatively small, so ε/C_n is non-negligible, $\theta_n = 1 - \varepsilon/C_n$ is meaningfully less than 1, and $\eta_{\text{ind}} = C_n/(C_n - \varepsilon)$ is correspondingly large. For evolutionary learning at L_0: C_0 is lineage survival, far exceeding ε, so $\theta_0 = 1 - \varepsilon/C_0 \approx 1$ and $\eta_{\text{evo}} \approx 1$. Since $\theta_0 \approx 1 \gg \theta_n$, it follows that $1/\theta_0 \ll 1/\theta_n$, therefore $\eta_{\text{evo}} \ll \eta_{\text{ind}}$. The learning rate difference is not assumed; it is derived from the cost-of-failure difference established in Theorem 4.

An individual may establish an informed action that reduces U in one context but fails in others. Only informed actions that reduce U across sufficient variation in W survive the evolutionary test — the higher rigor threshold filters for robustness. The analogy captures this shared directional property: both processes favor what reduces U at the relevant scale. The timescale, the rigor threshold, the object being updated, and the mechanism all differ; the directional relationship to U is what the analogy preserves.

Theorem 4 (Encoding Permanence Scales with Rigor). The certainty threshold required to promote an informed action f to encoding level L_i is:

$$\theta_i = 1 - \frac{\varepsilon_{\text{acceptable}}}{C_i}$$

where C_i is the cost of failure at level L_i and $\varepsilon_{\text{acceptable}}$ is the maximum acceptable expected cost of an encoding error, expressed in the same units as C_i. For heritable encoding (L_0), C_0 is lineage survival, requiring near-certainty across the full distribution of environments. For active reasoning (L_n), C_n is the cost of a single wrong action, a far lower bar.

Proof. Encoding f at level L_i is irreversible at cost $r_i > 0$ (Definition 8). The expected cost of encoding an action with current confidence θ is:

$$\mathbb{E}[\text{encoding error cost}] = (1 - \theta) \cdot C_i$$

where $(1 - \theta)$ is the probability the action is wrong and C_i is the cost when it is. Promotion is safe when this expected cost does not exceed what is acceptable:

$$(1 - \theta) \cdot C_i \leq \varepsilon_{\text{acceptable}}$$

Rearranging:

$$\theta \geq 1 - \frac{\varepsilon_{\text{acceptable}}}{C_i}$$

The minimum threshold at which promotion is safe is therefore $\theta_i = 1 - \varepsilon_{\text{acceptable}}/C_i$. Below this threshold, the expected cost of encoding error exceeds $\varepsilon_{\text{acceptable}}$ and continued active engagement is preferred. At or above it, encoding is the lower-expected-cost choice. □

Consequence. You must be very certain to permanently change the source code. A bad encoding at L_0 does not kill one individual; it kills the lineage. The rigor of the test is proportional to the permanence of the consequence.

Note on error propagation through the hierarchy. A potential objection is that the proof derives θ_i for each level independently, whereas errors in a hierarchy compound: a failure at L_0 propagates upward. This objection assumes a naive cascading model in which errors at L_i arrive at L_{i+1} at rate $(1 - \theta_i)$ regardless of whether L_i could address them. That is not the escalation architecture.

Under escalation (Theorem 2), errors partition into two classes at each level: **handled errors** — failures L_i detects and resolves within its scope, which never reach L_{i+1} — and **unhandled exceptions** — failures outside L_i's domain or beyond its capacity to resolve, which escalate. The rate seen by L_{i+1} is not the compounded Bernoulli rate from below. It is the rate of genuinely irreducible errors at L_i: errors that L_i was correct not to handle, because the knowledge or capacity to resolve them does not exist at L_i. This is a different quantity from $(1 - \theta_i)$, and it is bounded by the design of θ_i itself — the threshold ensures L_i handles everything within its cost-justified scope.

Furthermore, when an unhandled exception escalates, the context at L_i was already disrupted by the failure: the action failed, and L_i's current trajectory is suspended or invalid. The attention cost at L_{i+1} is not paid on top of productive work; it is paid instead of broken work. Escalation is correct triage. The thresholds θ_i are therefore independent by construction: each level's $\varepsilon_{\text{acceptable}}$ is its own irreducible error domain, not an accumulation from below. The proof applies at each level without modification.

Definition 10 (Certain). An informed action f is **certain** when $U(w, K_f) \approx 0$ across sufficient variation in W to meet the promotion threshold θ_i (Theorem 4). Certainty is asymptotic: decreasing density of uncertainty until active engagement no longer justifies the residual gradient.

Corollary 6 (Bounded Adaptation). Certain informed actions escape the active context window permanently, freeing L_n for the next frontier of uncertainty.

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7. The Higher-Order Function Space

Definition 11 (Higher-Order Function Space). Let M be a function space operating on the joint (F, K) pair [Thrun1998]:

$$M : (F, K) \times \mathcal{L} \to (F, K)$$

where $\mathcal{L}$ is a learning signal (experience, execution results, selection pressure). M is the mechanism by which (F, K) changes — categorically different from F, not a more certain version of it.

M's type signature is complete without enumerating its internal structure, because M is the cross product of all dimensions that have not yet precipitated out at the current scale of the entity:

$$M = \prod_{i \in \text{remaining}} D_i$$

where the product ranges over all dimensions — drive, meta-learning rate, exploration policy, convergence criteria, and whatever else has not yet reached the scale threshold to become a named dimension. F, K, and $\mathcal{A}$ are factors that have already been extracted from this product. M is always the remainder: the undifferentiated joint space of everything not yet named. As the entity scales, additional dimensions precipitate out of M by the same normalization argument that precipitated K from F: when the shared structure of a dimension across M-instances is large enough that factoring it out reduces retrieval cost from $O(n \cdot k)$ to $O(n + k)$, selection pressure forces the extraction. Each precipitation shrinks M by one factor and names a new dimension.

The hierarchy is open upward: there is always potentially a $M^{(2)}$ that operates on $M^{(1)}$ the same way $M^{(1)}$ operates on (F, K). One level suffices until it does not — when the gradient can no longer be followed by improving (F, K) alone, pressure propagates upward. The required number of levels is determined by the problem, not prescribed by the framework. The precipitation argument applies at every level; the hierarchy is not a prescribed structure but the output of a single recursive process applied at increasing scale. See Open Question 7.

The degenerate cases of level merger generalize from the $F = K$ case: for any contiguous block of N precipitated dimensions together with the M residual, all $N + 1$ collapse to a single undifferentiated process when the domain is fully crystallized at all scales in that block. The merger condition is $U(w, K) \approx 0$ across the domain at all N scales simultaneously — one condition, not N pairwise conditions. Since M represents the full cross product of remaining dimensions, any merger including M collapses not just N named levels but the entire unprecipitated hierarchy simultaneously. $F = M$ is the maximally primitive state: action spans the full cross product of all remaining dimensions. $M = M^{(2)}$

means the cross product is self-similar under the meta-operation — evolution approaches this limit at the timescale of selection on selection mechanisms.

Definition 11a (M Efficiency). The efficiency η_M of M is the rate at which M improves the joint (F, K) capability per unit of learning signal:

$$\eta_M(E) = \mathbb{E}_{\mathcal{L}}\left[\frac{\max_{f' \in F'} G(f', K') - \max_{f \in F} G(f, K)}{|\mathcal{L}|}\right]$$

where $(F', K') = M((F, K), \mathcal{L})$ is the updated joint state after M acts, $G(f, K) = I(f\,;\, W \mid K)$ is the information gain of the best available informed action (Definition 3b) in bits, and $|\mathcal{L}|$ is the information content of the learning signal in bits: $|\mathcal{L}| = H(\mathcal{L})$, the entropy of the signal received. Both numerator and denominator are in bits; η_M is dimensionless, measuring improvement in best-available information gain per bit of learning signal. η_M is defined for $H(\mathcal{L}) > 0$. When $H(\mathcal{L}) = 0$, the learning signal is degenerate (constant; carries no information) and η_M is undefined: the entity received no learnable signal and the rate of improvement per bit is not defined.

η_M is a signed quantity ranging over $(-\infty, \infty)$. $\eta_M > 0$: M improves the best available action in expectation — the entity learns. $\eta_M = 0$: M cannot improve (F, K) on average — the entity executes without learning. $\eta_M < 0$: M degrades (F, K) locally — overfitting, catastrophic forgetting, destructive updating, or deliberate exploration. An entity with *persistently* $\eta_M < 0$ does not persist under selection pressure. An entity that can act on $\eta_M < 0$ *temporarily* — when local gradient following would trap it — has strictly higher long-run persistence probability than one constrained to $\eta_M \geq 0$ at every step. η_M is operationally independent of $\mathcal{A}$: one measures the mechanism of improvement; the other measures the drive to engage it.

On measurement. η_M as defined requires computing $G(f, K) = I(f\,;\, W \mid K)$, which depends on the true joint distribution $P(W, \phi(f, w))$ — the distribution the entity is trying to learn. The entity cannot compute its own η_M exactly. This is not a defect of the definition but a structural feature: η_M is a property of the entity-world coupling, not a quantity the entity accesses directly. The same is true of $\mathcal{A}$, whose operationalization (Definition 6) requires varying survival pressure externally. Both quantities are measurable by an observer with access to the entity's trajectory and outcomes — the standard setting for evaluating learning systems — but not by the entity itself in real time. The theory's predictions (Theorems 1, 2, 4, 6) do not require the entity to compute $I(E)$; they follow from the structural constraints regardless of whether the entity knows its own intelligence.

Remark on exploration and the necessity of acting on negative $I(E)$. Theorem 3 Part II establishes that $\mathcal{T}_{reachable}$ grows without bound: the gradient landscape is non-stationary and non-convex. An entity constrained to $\eta_M \geq 0$ at every step is a pure gradient follower — it converges to the nearest local optimum

and remains there. But local optima become maladaptive as new uncertainty domains emerge: an entity trapped at a local optimum will eventually encounter $U > U^d_{lethal}$ in a domain it cannot reach from where it is. Selection removes it.

Therefore temporarily negative η_M — locally irrational steps to escape local optima — is survival-optimal when persistence requires $\mathbb{E}_t[\eta_M] > 0$ over the trajectory, not $\eta_M \geq 0$ at every step. Exploration and creative destruction are the signature of an M sophisticated enough to optimize over trajectories rather than steps.

M is itself subject to gradient descent at the meta-learning timescale, with the same rigor thresholds (Theorem 4). F acts on W; M acts on (F, K).

Theorem 5 (Type Irreducibility). No element of F can become an element of M through any learning process, regardless of certainty achieved.

Proof. $F : K \times W_{obs} \to \Delta A$. $M : (F, K) \times \mathcal{L} \to (F, K)$. These functions have different type signatures and operate over different domains. No learning process changes the type of a function. An informed action $f \in F$ that achieves certainty is promoted to a lower encoding level; it does not change type. M takes (F, K) as input and returns an updated (F, K); f takes knowledge and observations as input and returns an action distribution. These are irreducibly distinct. □

Corollary 5a (Distinct Roles). F navigates W given K. M evolves (F, K). Neuroplasticity is M acting on F; insight is M acting on K; habit formation is M promoting from L_n to L_2; evolution is M acting on M itself at the highest rigor threshold.

Theorem 6 (Necessary Conditions for Persistence: $\mathcal{A}$ and η_M as Structural Invariants). For any entity E persisting in W under selection pressure U^d_{lethal} with bounded context C_n, in a world with positive entropy (one that generates novel configurations over time):

1. $\mathcal{A}(E) > 0$ necessarily
2. $\eta_M(E) > 0$ necessarily
3. Every other scalar property of E can equal zero for some persistent entity
4. Therefore $I(E) = \mathcal{A} \cdot \eta_M$ is the natural measure of persistence — identified by what survival requires, not stipulated

Scope note. This theorem establishes that $\mathcal{A} > 0$ and $\eta_M > 0$ are *necessary conditions* for persistence. The proof is a selection argument: it shows that entities lacking either property are eliminated. This explains why $\mathcal{A} > 0$ entities predominate — it does not explain the *origin* of $\mathcal{A}$, i.e., how self-maintaining structures with $\mathcal{A} > 0$ bootstrap from initial conditions where no such structures exist. The origin question is a distinct problem, addressed as an open conjecture in Section 11 (Open Question 1). The framework takes persistence as its starting analytical condition and derives structural requirements from it; it does not claim to explain why anything persists in the first place.

Proof.

Part 1 (Necessity of $\mathcal{A} > 0$). Suppose $\mathcal{A}(E) = 0$. By Definition 6, E does not respond to gradient signals from W: $(F(t), K(t)) = (F(0), K(0))$ for all t. W is a physical state space with positive entropy: it generates novel configurations that no finite $K(0)$ anticipates. This is not a modeling assumption but a thermodynamic fact — any W with energy gradients produces states not represented in a fixed, finite knowledge graph. Since $K(0)$ is fixed and was not constructed with these novel states in mind, $U(w, K(0))$ is unconstrained there.

The selection argument does not require that any single entity encounters lethal novelty within its lifetime. It operates on lineages. An entity with $\mathcal{A} = 0$ reproduces copies of itself — copies that inherit the same fixed $(F(0), K(0))$. Those copies disperse across W. In a world with positive entropy, the lineage's collective coverage of W grows with each generation, even though each individual's (F, K) is static. The lineage therefore encounters the novel configurations that W generates: states where $U(w, K(0)) > U^d_{lethal}$ (Definition 5a). No copy can adapt ($\mathcal{A} = 0$); copies in those regions are removed. Copies in stable niches persist temporarily, but the niches themselves are not permanent — W has positive entropy, so no region stays static indefinitely. The $\mathcal{A} = 0$ lineage contracts as its environment shifts until it is eliminated. Selection removes the lineage. Any lineage that persists must contain entities that update (F, K) — i.e., entities with $\mathcal{A} > 0$.

The mechanism by which $\mathcal{A} > 0$ is enforced is timescale-specific. At the individual learning timescale: entities with $\mathcal{A} = 0$ do not engage M on gradient signals and are directly removed. At the evolutionary timescale: lineages that fail to produce offspring with $\mathcal{A} > 0$ are eliminated through differential reproduction — no individual requires $\mathcal{A}$ explicitly; the population-level filter enforces the property on lineages. At the cultural timescale: groups that do not maintain the institutional equivalent of $\mathcal{A}$ (the drive to reduce shared uncertainty) are outcompeted. The necessity is universal; the enforcement mechanism is timescale-specific. □

Part 2 (Necessity of $\eta_M > 0$). Suppose $\mathcal{A}(E) > 0$ but $\eta_M(E) \leq 0$.

Case $\eta_M = 0$: E has the drive to reduce U but M produces no net improvement in (F, K) per unit of learning signal: the best available action's information gain does not increase after M acts. (F, K) is effectively fixed despite $\mathcal{A}$. This reduces to Part 1's condition: with (F, K) fixed, the frontier of $\mathcal{T}_{reachable}$ grows beyond K's coverage. Additionally, C_n is bounded (Definition 8a). As E encounters novel world states, L_n accumulates unresolved context. By Theorem 2a, inference quality ρ degrades as L_n fills without promotion of resolved actions to lower encoding levels. Promotion requires $\eta_M > 0$: Theorem 4 establishes that encoding at level L_i requires rigor threshold $\theta_i = 1 - \varepsilon/C_i$, and achieving that threshold requires M to improve the action to the required certainty. With $\eta_M = 0$, no action ever reaches its threshold: L_n is never relieved, $\rho \to 0$, and inference eventually fails. U exceeds U^d_{lethal} in the domains that required the new actions. Selection removes E.

Case $\eta_M < 0$ persistently: M degrades (F, K) on average across the trajectory. This accelerates the failure of the $\eta_M = 0$ case: (F, K) worsens in expectation, U approaches U^d_{lethal} from above. Selection removes E strictly faster. Note that temporary $\eta_M < 0$ — deliberate exploration to escape local optima — is compatible with persistence and can increase long-run survival probability (see Remark in Definition 11a). The persistence condition is $\mathbb{E}_t[\eta_M] > 0$ over the trajectory, not $\eta_M(t) > 0$ at every step.

Therefore any $\mathcal{A} > 0$ entity that persists must have $\mathbb{E}_t[\eta_M] > 0$. □

Part 3 (Minimality: no other property is invariant). For every other scalar property of E, there exists a persistent entity for which it equals zero or its minimum value:

- $K(0) = \emptyset$*:* An entity with no initial knowledge survives if $\mathcal{A} > 0$ and $\eta_M > 0$ — it builds K from gradient descent, beginning with the first feedback signal received from W (Theorem 3, Part I).
- $F(0) = \{id\}$*:* F is a function space; its minimal element is the identity — the action that changes nothing. The identity element exists independently of K: it requires no grounding because it makes no knowledge claim. Executing the identity against W still returns feedback — the world observed in the absence of action is itself a signal. That feedback grounds the first proposition $p \in K$, after which the first non-trivial informed action can be constructed. The minimally non-empty F is therefore $\{\text{id}\}$, not $\emptyset$, and it is sufficient to begin gradient descent.
- $C_n = C_{min}$*:* An entity with the smallest viable context window survives if the graph structure (Theorem 2b) keeps $\rho = 1$ — C_n's absolute size is not invariant, only its boundedness.
- *Substrate, sensing σ, world projection W_{obs}:* These are free parameters of Definition 4 and Definition 1. The theory is parametric in all of them; no specific value is required for persistence. Entities on radically different substrates (biological, computational, collective) can all satisfy the survival constraint.

No scalar property other than $\mathcal{A}$ and η_M has the survival-critical necessity shown in Parts 1 and 2. □

Corollary 6a ($I(E) = \mathcal{A} \cdot \eta_M$ is structurally necessary, not stipulated). From Theorem 6, $\mathcal{A}$ and η_M are the unique invariants of persistence under bounded context and selection pressure. Any measure of intelligence must be positive if and only if both are positive, and zero if either is zero.

The product is not a choice — it is the only normalized combination of a probability and a conditional rate. $\mathcal{A} \in [0, 1]$ is the probability that E engages M in response to a gradient signal. η_M is the rate of improvement conditional on M being engaged. By the law of total expectation:

$$\mathbb{E}[\text{improvement rate}] = \mathcal{A} \cdot \eta_M + (1 - \mathcal{A}) \cdot 0 = \mathcal{A} \cdot \eta_M$$

Any other combination fails: $\mathcal{A} + \eta_M$ assigns positive intelligence to a zero-drive entity (contradicting Theorem 6 Part 1); $\min(\mathcal{A}, \eta_M)$ erases the independent invariance of Parts 1 and 2. The product is the same structure as Theorem 1's retention criterion: probability of needing it times conditional value.

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8. The Unified Model

Definition 12 (Trajectory). A trajectory τ is a sequence of informed actions drawn from F:

$$\tau = (f_1, f_2, \ldots, f_n), \quad f_i \in F$$

The **performance** realized by following trajectory τ is:

$$P(E, \tau) = \mathbb{E}_W \left[U(w_0, K_0) - U(w_n, K_n) \right]$$

This is the total uncertainty reduction achieved — a measure of what current (F, K) accomplished, not of intelligence as defined. $P(E, \tau)$ grows with trajectory length and with the quality of F at the time of execution. Intelligence $I(E) = \mathcal{A} \cdot \eta_M$ is what determines how quickly P improves across successive trajectories: an entity with high $I(E)$ builds better (F, K) between trajectories, so P grows faster over time than for an entity with low $I(E)$ starting from identical initial conditions.

Agency manifests as a trajectory τ: a sequence of informed actions selected and executed over time. Each action reduces U, returns a learning signal to M, and M uses that signal to improve (F, K) for the next action. Intelligence is not visible in any single step; it is the rate at which the steps improve.

Conclusion

The formal results establish a minimal structural account of intelligence under bounded context. The account rests on four primitives: a world state space W generating uncertainty, a bounded active context C_n, a survival threshold U^d_{lethal}, and a growing reachable space $\mathcal{T}_{reachable}$. From these:

- Knowledge and action are inseparable: K is constitutively indexed by F, with an asymmetric retention criterion derived from survival pressure (Theorem 1).
- Graph-structured K is necessary: bounded context requires progressive disclosure, which requires sub-linear retrieval, which requires stored adjacency — a graph ([McCarthy2026a]).
- Escalation dominates top-down allocation: bottom-up failure signaling is $O(1)$ in steady state; top-down precision allocation degrades as $O(|F|/C_n)$ (Theorem 2a).

- The encoding hierarchy is a necessary consequence of bounded C_n: proven knowledge promotes to lower levels, freeing the active context for genuine novelty (Theorem 4).
- Agency $\mathcal{A} > 0$ and learning efficiency $\eta_M > 0$ are structural invariants of any persisting entity under selection pressure (Theorem 6).
- Intelligence $I(E) = \mathcal{A} \cdot \eta_M$ is the driven capacity to improve — not what has been learned, but how efficiently evidence converts into better (F, K).

The framework is substrate-independent: it specifies structural requirements, not implementation. Any system satisfying these constraints — biological, artificial, or hybrid — is a member of the structural class. The biological architecture is accommodated as one instance: heritable encoding at L_0, individual learning at L_n, and cultural transmission as external K when the heritable channel cannot carry individual (K, F).

The theory makes testable predictions: that graph structure emerges under bounded-context optimization (confirmed in the engineering system that motivated the framework); that top-down orchestration degrades under scaling (observable in both software and cognitive architectures); and that the encoding hierarchy follows from cost structure rather than requiring separate explanation.

Extensions to collective intelligence, superintelligence, and sentience — including formal definitions and convergence results — are developed in a companion paper.

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9. Related Work

Graph Necessity and Autonomous Learning

The graph necessity result on which this paper builds is established in [McCarthy2026a]: any system accumulating propositions under bounded active context must maintain a directed graph with typed edges. The proof proceeds from two independent arguments — information preservation (factoring is the only lossless space-creating operation, and factoring is graph construction) and retention dynamics (sub-linear dependency queries require stored adjacency). That paper also shows that continuous-weight retrieval mechanisms (e.g., transformer self-attention) implement graph structure rather than constituting an alternative to it.

Dupoux, LeCun, and Malik [DupouxLeCunMalik2026] propose a dual-system architecture (observation + action) with meta-control routing, motivated by cognitive science. Their framework is architectural and prescriptive: it argues that autonomous learning *should* integrate observation, action, and meta-control. The present work is complementary: rather than proposing what intelligent systems should contain, we derive what they *must* contain. Their Systems A and B correspond to F (informed actions on W) and K (observational propositions about W); their System M corresponds to the meta-control aspect of

M in this framework. The structural results here — type irreducibility of F and M (Theorem 5), the necessity of $\mathcal{A}$ and η_M (Theorem 6), and escalation dominance (Theorem 2a) — provide formal grounding for architectural choices that [DupouxLeCunMalik2026] motivates empirically.

Free Energy Principle and Active Inference

The Free Energy Principle [Friston2010] establishes that any self-organizing system that maintains its existence necessarily minimizes variational free energy — formally equivalent to bounding uncertainty about hidden world states. Active inference [Friston2017] extends this to action selection: the organism selects actions that minimize expected free energy, combining epistemic value (information gain) and pragmatic value (goal proximity).

Containment, not contradiction. This framework does not contradict FEP. FEP agents are members of this framework's structural class: they satisfy $\mathcal{A} > 0$ (self-organizing systems act), $\eta_M > 0$ (generative model improves over time), bounded C_n (inference under resource limits), and a survival condition: maintaining a Markov blanket (statistical separation of internal from external states) is the formal characterization of what it means for an entity to persist — the same structural requirement as $U < U^d_{lethal}$ stated in the language of conditional independence rather than uncertainty magnitude. Both describe maintained boundary; neither implies the other's formalism, but they pick out the same class of persisting systems. The variational free energy $\mathcal{F}$ decomposes into accuracy and complexity terms; the accuracy term $-\mathbb{E}_q[\log P(o \mid m, \pi)]$ is precisely the expected surprisal $H(W \mid K)$ of Definition 5. FEP agents therefore instantiate this framework with $U(w, K)$ realized as variational free energy and M realized as Bayesian belief updating. The claim is not that FEP is wrong. The claim is that it is a special case within a broader structural class — in the same way that Newtonian mechanics is not wrong but is contained within the class of systems satisfying more general symmetry constraints.

What the framework adds is four things FEP does not provide. First, FEP derives the minimization behavior from the requirement of self-organization but does not establish the minimization *imperative* — why $\mathcal{A} > 0$ is structurally necessary for persistence. Theorem 6 establishes this as a necessary condition: any entity that does not reduce uncertainty eventually fails and is removed, so $\mathcal{A} > 0$ is a structural invariant of persisting systems. (The origin of $\mathcal{A}$ — how self-maintaining structures bootstrap from initial conditions — is a separate open question; see Section 11.) Second, FEP's generative model is unconstrained in structure; we derive a structural constraint from survival pressure: K is necessarily indexed by F, with an asymmetric retention criterion that makes specific testable predictions about what knowledge is retained and what is pruned. Third, FEP's precision allocation model is top-down: higher levels direct attention to lower levels by weighting prediction errors. This is one specific implementation of M. We show formally (Theorem 2a) that this implementation degrades inference quality as capability grows under bounded context, because delegation

overhead fills C_n proportionally to $|F|$. The escalation architecture — bottom-up failure signaling with promoted crystallization — is a different implementation of M that is $O(1)$ in steady state. This is a critique of FEP's architectural *prescription* under bounded context, not of its minimization objective or its descriptive correctness. FEP's precision weighting may be optimal when C_n is not the binding constraint; under bounded context, escalation dominates.

Information-Theoretic Accounts

Shannon's mutual information [Shannon1948] grounds Definition 5 ($U = H(w \mid K)$) and Definition 3b ($G(f, K) = I(f\,;\, w \mid K)$). The data processing inequality [Cover2006] grounds the $G \geq 0$ bound. The information bottleneck principle [Tishby2011] treats the retention threshold for knowledge-action coupling as a design parameter; this paper derives that threshold from survival pressure, making it a consequence rather than a choice. Schmidhuber's formal theory of intrinsic motivation [Schmidhuber2010] formalizes curiosity as compression progress — the rate at which a learning algorithm improves its world model — which is structurally η_M. The difference: Schmidhuber treats the drive as a design objective; here $\mathcal{A} > 0$ is established as a necessary condition of persistence (Theorem 6), and η_M is separated from it as a formally independent quantity.

Cognitive Architectures

Global Workspace Theory [Baars1988, Dehaene2011] holds that local processors handle routine computation; the global workspace broadcasts only when local processors fail. This is the escalation architecture in cognitive science language. The contribution here: the escalation criterion is derived from U-reduction cost (Theorem 2) rather than assumed architecturally, the hierarchy is continuous rather than binary, and Theorem 2a shows why top-down allocation from the global workspace is inefficient. Kahneman's System 1/System 2 distinction [Kahneman2011] is the popular form of the same binary — corrected here by the continuous $\mathcal{E}(c, r)$ parameterization of Definition 8. Gibson's affordances [Gibson1979] are action possibilities indexed by organism capacities — K indexed by F under a different name. Theorem 1 provides what Gibson's account lacks: a formal selection criterion for which affordances are retained. Anderson and Schooler [AndersonSchooler1991] show empirically that human memory retention mirrors the statistical structure of the environment — precisely the prediction of Theorem 1's retention criterion applied to biological K.

Formal Theories of Intelligence

Hutter's AIXI [Hutter2005] defines the optimal agent as one that maximizes expected reward over all computable environments, using Solomonoff induction for prediction. Legg and Hutter [LeggHutter2007] derive from this a formal definition of intelligence as average performance across all computable environments weighted by Kolmogorov complexity. The present framework differs in three respects. First, AIXI assumes unbounded computation; the central constraint

here is bounded C_n and the consequences that follow from it. Second, AIXI defines intelligence as performance; this framework defines it as the rate of improvement (η_M), which is independent of current performance level. Third, AIXI's environment class is all computable functions; here the environment is physical (W with positive entropy), and the survival threshold U^d_{lethal} is a structural feature of the entity-world coupling, not a design parameter.

Chollet [Chollet2019] defines intelligence as skill-acquisition efficiency over novel tasks — structurally close to η_M as defined in Definition 11a. The distinction: Chollet treats the prior knowledge and experience of the agent as a confound to be controlled; here prior K is constitutive of intelligence because η_M operates on (F, K). Chollet's "scope" and "generalization difficulty" map to the growth of $\mathcal{T}_{reachable}$; his "priors" correspond to the encoding hierarchy's lower levels. The frameworks are compatible but ask different questions: Chollet asks how to measure; this paper asks what structural conditions are necessary.

Bounded Rationality and Resource-Rational Analysis

Simon [Simon1955] established that rational agents operating under computational and informational constraints do not optimize globally but satisfice — selecting the first option that exceeds a threshold. The bounded context constraint C_n formalizes one specific source of the bounds Simon identified. The retention criterion (Theorem 1) is a satisficing criterion: retain p if benefit is not certainly zero, rather than computing the globally optimal K.

Griffiths, Lieder, and Goodman [GriffithsEtAl2015] formalize resource rationality: the optimal algorithm given computational constraints may differ from the optimal algorithm given unlimited resources. This is precisely the argument of Theorem 2a — top-down precision allocation is optimal without resource constraints; under bounded C_n, escalation dominates. The encoding hierarchy (Definition 8) is a resource-rational architecture: it allocates costly active inference only to genuinely uncertain problems, handling resolved problems at lower cost.

Clark [Clark2015] synthesizes predictive processing with embodied cognition, arguing that the brain is fundamentally a prediction engine that minimizes prediction error hierarchically. This is the cognitive science framing of what Theorem 2 formalizes: hierarchical error minimization under bounded context. Clark's "action-oriented predictive processing" corresponds to the K-indexed-by-F structure of Theorem 1 — predictions are constitutively tied to actions.

Pearl [Pearl2009] provides the formal framework for causal reasoning in directed graphs. The knowledge graph K as defined here (Definition 2) encodes inferential and causal relationships as a directed graph; the Bayesian network interpretation of Definition 5 is grounded in Pearl's framework. Pearl's do-calculus formalizes interventional reasoning — the distinction between observing and acting — which corresponds to the distinction between K (what is known) and F (what can be done). The retention criterion of Theorem 1 can be read as a causal relevance

filter: retain p if there exists a causal path from p to some $f \in F$ that reduces U.

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10. Discussion

The formal results have implications beyond the immediate theorems.

The encoding principle. Corollary 6 implies that as K grows, informed actions are promoted down the encoding hierarchy. Work that is currently active reasoning (uncertain, costly, engaged at L_n) becomes reflex, then encoding, then crystallized. By Theorem 2 and Definition 8, this frees the active context window for the next genuine frontier of uncertainty. $\mathcal{A}$ does not converge when its current domain is solved — it reaches outward.

The civilization-scale knowledge system as empirical evidence. An internal-model account predicts decelerating knowledge growth: larger models contain more variance, ρ degrades, and recovery requires expanding capacity (Theorem 2a). Civilizational knowledge growth has accelerated. The explanation: intelligence externalized K into graph-structured repositories and retrieves $K^{[f]}$ into active context per problem. This is the escalation architecture (Theorem 2) and graph necessity result ([McCarthy2026a]) at civilizational scale. Two predictions an internal-model account cannot make: (i) knowledge growth accelerates as K grows (Corollary 6 and Theorem 4: proven actions promote to lower levels, freeing L_n for new learning), and (ii) the external architecture self-assembles as a graph with provenance. Both are observed.

Limits of the formal account. The theory establishes structural requirements for individual intelligence under bounded context. It does not address collective intelligence, convergence to general intelligence, or the conditions under which shared knowledge reduces uncertainty faster than any individual. These extensions are developed in a companion paper.

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11. Open Questions

1. **The origin of $\mathcal{A}$ (distinct from its necessity).** Theorem 6 establishes that $\mathcal{A} > 0$ is a necessary condition for persistence: it is a structural invariant of any entity that survives under selection pressure with bounded context. This is a selection argument — it explains why $\mathcal{A} > 0$ entities predominate, not how $\mathcal{A} > 0$ structures arise from initial conditions where no such structures exist. The origin question is logically independent of the necessity result and remains open.

   The question: what mechanism produces the first self-maintaining structures with primitive $\mathcal{A} > 0$ from $\mathcal{A} = 0$ initial conditions? Can $\mathcal{A}$ be constructed, or only selected for once it exists?

A conjectured answer (not a formal result). $\mathcal{A}$ may be thermodynamically instantiated. $U(w, K) = H(w \mid K)$ is information entropy. Reducing U is reducing entropy locally. In any universe with entropy gradients, structures that maintain themselves against thermodynamic pressure persist preferentially. The first self-maintaining structure was, in the minimal sense, bounding its own uncertainty: $\mathcal{A}$ in primitive form. Not designed. A consequence of the physics of a world that destroys things.

The deeper question is therefore: is agency a natural consequence of the universe? The conjectured answer is yes: $\mathcal{A}$ is the thermodynamic attractor of any entropy-gradient-rich world given sufficient time. Wherever entropy gradients exist and structures can form, the structures that persist are those that resist dissolution by reducing local uncertainty. Given enough time, those structures compound. Intelligence is what that compounding produces.

Formal derivation of this claim from thermodynamic first principles (connecting U^d_{lethal} to physical entropy thresholds and demonstrating the inevitability of primitive $\mathcal{A}$ from dissipative system dynamics) remains open. See [Prigogine1984] (dissipative structures), [England2013] (dissipative adaptation), and [Schrodinger1944] (negative entropy and life) for adjacent physical frameworks.

2. **The direction of $\mathcal{A}$.** Is agency a uniform drive (a general orientation toward uncertainty reduction), or does it carry direction? Survival pressure selects for reducing *specific* uncertainties. Does $\mathcal{A}$ encode this specificity, or does F and K direct it toward particular regions of W after the fact?

3. **Perception and K construction.** How does σ determine which features of W_{obs} are encoded into K? Is σ itself subject to gradient descent, and if so, at what level?

4. **Bound composition.** Can bounds compose, i.e., can $f_1 \circ f_2$ produce informed actions covering regions of W neither f_1 nor f_2 could cover alone? What are the rules of composition?

5. **The novelty boundary (theory limit).** The framework defines informed action over regions of W where K provides grounding. At $U(w, K) \approx 1$ — where no existing informed action applies — the theory does not specify E's behavior. This is a boundary condition, not a design choice: the model is silent on uninformed action by construction. Behavior in this regime (random exploration, interpolation, escalation to M) is outside the formal scope and would require extension of the framework.

6. **Competing bounds.** When multiple bounds $f_i \in F$ are applicable to the same region of W, how does E select among them? Is there a selection function, and is it itself in F or M?

7. **The $M^{(n)}$ hierarchy.** The framework defines M as operating on (F, K) (Definition 11) and notes that the same type applies recursively. The

formal questions are: how many levels are required, what determines when another level becomes necessary, and whether the hierarchy has a natural termination point?

The framework's answer to the first question is empirical, not prescribed: one level of M suffices until it does not. As long as the gradient on (F, K) can be followed by improving (F, K) directly, $M^{(1)}$ is adequate. When it cannot — when the learning mechanism itself is the bottleneck — selection pressure propagates to $M^{(2)}$. A single level is fine, until it isn't. The required depth is determined by the structure of the problem and the selection pressure it generates, not by the theory in advance. The author's computational system has two explicit levels; there is no formal argument that two is sufficient in general.

Hypothesis (M-depth driven by selection pressure). Each level $M^{(n)}$ is subject to the same minimization imperative as the level below: it will be displaced or refined if a better $M^{(n+1)}$ exists and the selection pressure is sufficient to instantiate it. The hierarchy has a natural termination at level n^* only if the gradient at that level reaches zero — no available $M^{(n^*+1)}$ can improve the learning mechanism further given current W. Whether such a fixed point exists, and whether it is reachable in finite time under realistic selection pressure, is open. Evolution [Hinton1987] and meta-learning [FinnEtAl2017, Thrun1998] are empirical candidates for $M^{(2)}$; whether anything instantiates $M^{(3)}$ in known systems remains unclear.

8. **Genetic encoding and the developmental sufficiency requirement.** The encoding hierarchy (Definition 8) describes a ratchet: proven behaviors promote toward L_0 as they meet the rigor threshold $\theta_0 \approx 1$ (Theorem 4). The question is what the asymptotic product of evolutionary gradient descent encodes [Schrodinger1944, MaynardSmith1995, Hinton1987, England2013].

   The framework does not resolve the content of genetic encoding — what specific mixture of M, (F_0, K_0), and developmental constraint DNA carries across all levels is an open empirical question. The claim is narrower: whatever genetic encoding carries must be *sufficient to develop the entity* — sufficient to bootstrap an organism capable of building (F, K) through interaction with W. The model is compatible with genetic encoding at any level of the hierarchy, in any proportion, provided the developmental output is a system that can run the gradient.

   What the framework does predict about the evolutionary rigor threshold: specific (F, K) are environment-dependent — a skill that reduces U in one environment may not transfer across the full distribution of environments the lineage will encounter. Content that clears θ_0 must generalize across that distribution. Developmental machinery — the programs, architectures, and gradient signals that build (F, K) — is more likely to generalize

than specific content, because it is environment-agnostic. The framework predicts selection pressure favoring developmental generality over content specificity at the highest rigor levels; it does not predict that specific (F_0, K_0) is absent.

This is consistent with the observation that we are not born with the context window of our parents: individual (K, F) — everything learned in one lifetime — clears θ_n but not θ_0. The inheritance mechanism transmits what is sufficient for development, not what was learned. The diversity of knowledge across individuals with similar genetic endowment is the expected outcome: the same developmental machinery operating on different environments produces different (K, F).

Cultural transmission (language, writing, institutions) provides a parallel channel for (K, F) at lower fidelity but higher speed — shared K functioning as external inheritance, serving a transmission function that heritable encoding cannot match for individual-lifetime content. The two channels operate at different rigor thresholds and different timescales.

Formal questions that remain open: the precise allocation between M-encoding and (F_0, K_0)-encoding across evolutionary lineages; whether the modularity of genetic regulatory networks is derivable from the type structure of $M : (F, K) \times \mathcal{L} \to (F, K)$ at the evolutionary scale; and what determines θ_0 in terms of environmental distribution statistics.

Further open questions concerning collective intelligence, the AGI condition, and the structure of genetic encoding are developed in a companion paper.